



Port 1)

$$x_1 \sim \mathcal{N}(0, \Sigma)$$

$$x_2 | x_1 \sim \mathcal{N}(A x_1, \Sigma)$$

$$x_2 | x_2 \sim \mathcal{N}(A x_2, \Sigma)$$

$$\begin{aligned}\Sigma_{11} &= \text{cov}(x_1, x_1) \\ &= \text{Var}(x_1) \\ &= \Sigma\end{aligned}$$

$$\begin{aligned}\Sigma_{12} &= \text{Cov}(x_1, x_2) \\ &= \mathbb{E}[(x_1 - \mathbb{E}(x_1))(x_2 - \mathbb{E}(x_2))]\end{aligned}$$

Q.1.1

$$p(x_1 | x_2) = \mathcal{N}(x_1 | \mu_{1|2}, \Sigma_{1|2})$$

$$X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \mathcal{N}\left(\mu = \begin{pmatrix} 0 \\ A x \end{pmatrix}, \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}\right)$$

Part 2

2.1 determine likelihood for the regression model in $y_n = f(x_n) + \epsilon_n$

- we have a following model: $y_n = f(x_n) + \epsilon_n$, where $\epsilon_n \sim \mathcal{N}(0, \sigma^2)$
- likelihood for a single-point: $p(y_n | f_n) = \mathcal{N}(y_n | f(x_n), \sigma^2)$
- likelihood for entire dataset: $p(y | f) = \mathcal{N}(y | f, \sigma^2 I)$

2.2 Determine the analytical prior prediction distribution $p(y^* | x^*=1)$ for $y^* = y(x^*)$

• $f^* := f(x^*) \rightarrow$ function value at test point

• $y^* := f^* + \epsilon^* \rightarrow$ noisy observation at test pt.

• from the GP Prior $f^* \sim \mathcal{N}(0, k(x^*, x^*))$

• from the noise model $\epsilon^* \sim \mathcal{N}(0, \sigma^2)$, independent of f^*

$$\therefore y^* = f^* + \epsilon^* \sim \mathcal{N}(0, k(x^*, x^*) + \sigma^2)$$

$$k = 0.7$$

$$\sigma = \frac{1}{2} \sqrt{2}$$

$$\sigma^2 = \frac{1}{2} = 0.5$$

$$p(y^* | x^*=1) = \mathcal{N}(0, k(x^*, x^*)) = \mathcal{N}(0, k(1, 1))$$

$$\Rightarrow k(x, x') = k^2 \exp\left(-\frac{1}{2k^2} \|x - x'\|_2^2\right)$$

$$\begin{aligned} k(1, 1) &= 0.7^2 \exp\left(-\frac{1}{2 \cdot \left(\frac{\sqrt{2}}{2}\right)^2} \|1 - 1\|_2^2\right) \\ &= 0.49 \exp\left(-\frac{1}{2 \cdot \frac{2}{4}} \|1 - 1\|_2^2\right) \end{aligned}$$

$$f^*(x=1) = 0.49 \exp\left(-\cancel{\|1 - 1\|_2^2}\right)$$

$$\begin{aligned} \text{to obtain } y^*(x=1) &= f^* + \sigma^2 \\ &= k^2 + \sigma^2 \\ &= 0.7^2 + 0.2^2 = 0.53 \end{aligned}$$

$$\therefore p(y^* | x=1) = \mathcal{N}(0, k^2 + \sigma^2) = \underline{\underline{\mathcal{N}(0, 0.53)}}$$

2.3 Determine the analytical post. pred. dist $p(y^*|y, x^*=227)$ for $y^*=y(x^*)$

- we start w/ prior on function values, $f \sim GP \sim (0, k(x, x))$, this means any finite collection of function values follow mult. Gauss.
- we add noisy observations, each observation y_n is: $y_n = f_n + \epsilon_n$, $\epsilon_n \sim N(0, \sigma^2)$, so the observed y vector is "noisy" version of f .
- posterior over test value $f^* = f(x^*)$: given observed y and the GP prior, the posterior over f^* is:

$$p(f^*|y, x^*) = \mathcal{N}(\mu_*, \sigma_*^2)$$

$\sigma_*^2 = k(x^*, x^*) - k_{*}^T (K + \sigma^2 I)^{-1} k_*$
 $\mu_* = k_{*}^T (K + \sigma^2 I)^{-1} y$
 $\leftarrow N \times N$ mat of train inputs
 $\leftarrow N \times 1$ ker. vector between test input x^* and training input X .

- add noise to get dist. over y^*

$\rightarrow$ we want $y^* = f^* + \epsilon$, $\epsilon \sim N(0, \sigma^2) \Rightarrow \therefore p(y^*|y, x^*) = \mathcal{N}(\mu_*, \sigma_*^2 + \sigma^2)$

2.4 Determine the analytical expression for the marginal likelihood $p(y|h, l, \sigma)$ and compute the num. val. $\log p(y|h, l, \sigma)$

- to find the marginal likelihood of the data $p(y)$, we need to marginalize f out of $\mathcal{N}(f, \sigma^2 I, l, l)$

$$p(y) = \int p(y|f) p(f) df$$

$$p(y) = \int \underbrace{p(y|f, \sigma^2 I)}_{\text{likelihood}} \underbrace{p(f|h, l)}_{\text{prior}} df$$

$$p(y) = \mathcal{N}(y|0, K + \sigma^2 I) = -\frac{1}{2} y^T (K + \sigma^2 I)^{-1} y - \frac{1}{2} \log |K + \sigma^2 I| - \frac{n}{2} \log(2\pi)$$

(this is just standard multivariate normal dist.)
 python: `np.linalg.logpdf(y, mean=0, cov=K + SIGMA**2 * np.eye(N))`

- using the function log-marginal-likelihood we get the value $\log p(y) = -6.65$

Question 2.5: Determine the analytical posterior distribution $p(f^*|y, x^* = 1)$ for $f^* = f(x^*)$ as well as the analytical posterior predictive distribution for $p(y^*|y, x^* = 1)$ for $y^* = y(x^*)$.

- we use the equations for finding the post predictive

$$p(f^*|y, x=1) = \mathcal{N}(f^*|0.07, 0.14)$$

$$p(y^*|y, x=1) = f^* + \epsilon = \mathcal{N}(y^*|0.07, 0.18)$$

Consider now a different kernel

$$k_2(x, x') = c_1 \left(1 + \frac{\|x - x'\|^2}{2\ell^2} \right)^{-1} + c_2 x x',$$

with hyperparameters $c_1, c_2, \ell > 0$.

Question 2.6: Determine whether k_2 is a stationary kernel and determine whether k_2 is an isotropic kernel.

- k_2 would be a stationary kernel, because we observe, that it depends on difference between two points $(x_i - x_j)$, it would also be isotropic, bcs. it has a norm measure, however the term $x \cdot x'$ makes it not being neither isotropic nor stationary.

Part 3: Binary classification

Consider the following generalized linear model for binary classification:

$$\begin{aligned} y_n | \mathbf{w}, x_n &\sim \text{Ber}(\sigma(f(x_n))) \\ f(x) &= w_0 + w_1 x + w_2 x^2 \\ \mathbf{w} &\sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \end{aligned}$$

where $\mathbf{w} = [w_0 \ w_1 \ w_2] \in \mathbb{R}^3$, $\mathbf{I} \in \mathbb{R}^{3 \times 3}$ is the identity matrix, $\sigma(\cdot)$ is the logistic sigmoid function, $x_n \in \mathbb{R}$ and $y_n \in \{0, 1\}$.

Assume $\hat{\mathbf{w}}_{\text{MAP}} = \begin{bmatrix} 2.647 \\ -1.688 \\ -0.596 \end{bmatrix}$ is the MAP-estimator for $\mathbf{w}$ given some dataset $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$, where $\mathbf{y} \in \mathbb{R}^N$ denotes the targets.

Question 3.1: Suppose you suspect a bug in the code for computing the MAP estimator. Explain how you could verify that $\hat{\mathbf{w}}_{\text{MAP}}$ is indeed the correct MAP estimator.

Question 3.2: Determine the posterior predictive distribution for $p(y^*|y, x^* = -3)$ using the plug-in approximation based on the MAP-estimator.

- we are asked to $p(y^*|y, x^* = -3)$. This is the post. pred. dist. for a new data pt. $x = -3$. We integrate over all possible parameter values:

$$p(y^*|y, x^*) = \int p(y^*|w, x^*) p(w|y) dw$$

- this translates to: average the prediction over ALL plausible weights vectors w , weighted by how likely they are given the data

- we are using the plug-in approximation, which replaces the full posterior:

$p(w|y)$ with a point mass at the MAP estimates:

$$p(w|y) \approx \delta(w - \hat{w}_{\text{map}})$$

- then the predictive becomes:

$$p(y^*|y, x^*) \approx p(y^*|\hat{w}_{\text{map}}, x^*)$$

and since we are using a logistic model, this is a Bernoulli distribution $w/$

$$p(y^*=1|\hat{w}_{\text{map}}) = \sigma(f(x^*))$$

where $f(x) = w_0 + w_1 x + w_2 x^2$

Writing it up: the predictive dist. under the plug-in approx is

$$p(y^*|y, x^*) \approx \text{Bernoulli}(y^*|\sigma(f(x^*)))$$

with: $f(x^*) = \hat{w}_0 + w_1 x^* + w_2 (x^*)^2$

we are given: $\hat{w}_{\text{map}} = \begin{bmatrix} 2.647 \\ -1.688 \\ -0.596 \end{bmatrix}$, $x^* = -3$ so $f(-3) = 2.347 \Rightarrow$ then $p(y^*=1|y, x^*=3) \approx \sigma(2.347) \approx \underline{\underline{0.91}}$

$\Rightarrow$ which tells us that $y^*=1$ w/ 91% probability

Recap

$$\bullet p(y^*=1|y, x^*) \approx \sigma(f(x^*))$$

To give an example of the challenge: At $x^*=-3$, we have 91% chance of explosion, and **ONLY** 9% that it won't explode.

$$\bullet p(y^*=0|y, x^*) = 1 - \sigma(f(x^*))$$

- we are given the plug-in map estimate $\hat{w}_{\text{map}} = \begin{bmatrix} 2.647 \\ -1.688 \\ -0.596 \end{bmatrix}$, we can plug these

- into the $f(x^*) = \begin{bmatrix} 1 \\ x^* \\ x^{*2} \end{bmatrix}^T$, to obtain the prediction for $x^*=-3$.

$$\therefore y^* = \underbrace{f(-3)}_{1 \times 3} \cdot \underbrace{w}_{3 \times 1} = 2.347$$

scalar

now we feed this through sigmoid

$$\sigma(2.347) = 0.91$$

$$\therefore p(y^*=1|x=-3) = 0.91$$

Question 3.3: Use the Gaussian approximation to compute a 90% posterior credibility interval for w_0 .

to compute 90% posterior credibility interval, we need to sample from the Gaussian, we use

$$\hat{w}_{MAP} = \begin{bmatrix} \hat{w}_0 \\ \hat{w}_1 \\ \hat{w}_2 \end{bmatrix} = \begin{bmatrix} 2.647 \\ -1.688 \\ -0.596 \end{bmatrix}, \quad S = \begin{bmatrix} 3 & -0.39 & -0.3 \\ -0.39 & 1.55 & 0.37 \\ -0.3 & 0.37 & 0.14 \end{bmatrix} \Rightarrow \text{Since we are looking for 90\% post. cred. interval for } w_0. \text{ We } \therefore \text{ do not need to sample from Multivariate, but just from Marginal } p(w_0|y) \approx \mathcal{N}(w_0 | 2.647, 3.00)$$

or, $W_{MAP}[0] = 2.647; \Sigma_{00} = 3.00$

we draw $N = 10^6$ samples from $\mathcal{N}(\mu_{MAP}[0], \Sigma_{00}) \Rightarrow w_0^{(i)} \sim \mathcal{N}(\mu, \sigma^2), i = 1, \dots, N$

```
Python:
mu = w_map[0]
sigma = np.sqrt[0]
logp = jax.random.PRNGKey(42)
samples = mu + sigma * jax.random.normal(logp, shape=(10**6,))
interval = jnp.percentile(samples, jnp.array([5, 95]))
```

estimate the 90% cred. int. as the empirical 5th and 95th percentiles of the samples:

$$CI_{90\%}(w_0) \approx [Quantile_{0.05}, Quantile_{0.95}]$$

$$\mu = \hat{w}_0 = 2.647$$

$$\sigma^2 = 3 \Rightarrow \sqrt{3} \approx 1.732$$

A 90% posterior credibility interval under the normal distribution is:

$$[\mu - 2.05 \cdot \sigma; \mu + 2.05 \cdot \sigma]$$

0 $Z_{0.05}$ is a quantile for 90% central mass (5% in each tail)

from stand. norm. tables: $Z_{0.05} \approx 1.645$

- plugging in

$$[2.647 - 1.647 \cdot 1.732; 2.647 + 1.645 \cdot 1.732]$$

$$[-0.201; 5.495]$$

$$\therefore p(w_0 \in [-0.20, 5.49] | D) = 0.90$$

Question 3.4: Use the Gaussian approximation to compute the approximate posterior distribution $p(f^*|y, x^* = -3)$ and the approximate posterior predictive distribution for $p(y^*|y, x^* = -3)$ using the probit approximation.

- for inputs x , we have polynomial features

$$\phi(x) = \begin{bmatrix} 1 & x & x^2 \end{bmatrix}^T \in \mathbb{R}^3 \text{ weights}$$

and the logistic link

$$p(y=1|x, w) = \sigma(w^T \phi(x)), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

with a standard prior $w \sim \mathcal{N}(0, I)$ and data (X, y) , the posterior is intractable. We approximate it by Gaussian Laplace:

$$p(w|y) \approx q(w) = \mathcal{N}(w|m, S) \quad \text{inverse of a Hessian matrix}$$

$m \propto W_{MAP}$

From weights to the latent f^* at a new input

- for a new input x^* define the feature vector $x^* = \phi(x^*) \in \mathbb{R}^3$ and the latent score

$$f^* := w^T x^*$$

Under the Gaussian approximation for w , the linear-Gaussian transform implies

$$f^*|y, x \sim \mathcal{N}(m^*, v^*), \quad m^* = m^T x^*, \quad v^* = x^{*T} S x^*$$

Posterior Predictive as an expectation over f^*

$$p(y^*=1, y, x^*) = \int \sigma(w^T x^*) p(w|y) dw \approx \int \sigma(f^*) \mathcal{N}(f^*|m^*, v^*) df^* = \mathbb{E}_{f^* \sim \mathcal{N}(m^*, v^*)} [\sigma(f^*)]$$

To solve 3.4

let $\phi(x) = \begin{bmatrix} 1 \\ x \\ x^2 \end{bmatrix}$, $f(x) = w^T \phi(x)$, $\sigma(z) = \frac{1}{1 + e^{-z}}$

Under Gauss approx: $p(w|y) \approx \mathcal{N}(w|w_{MAP}, S)$, the latent at a new input x^* is linear in w , hence Gauss

$$p(f^*|y, x^*) = \mathcal{N}(f^*| \mu_F, \sigma_F^2)$$

with $\mu_F = \phi(x^*)^T w_{MAP}$, $\sigma_F^2 = \phi(x^*)^T S \phi(x^*)$

For $x^* = -3$, we have $\phi(x^*) = [1, -3, 9]^T$. Using $w_{MAP} = (2.647, -1.688, -0.596)^T$ and the given S

$$\mu_F = [1, -3, 9]^T w_{MAP} \approx 2.347, \quad \sigma_F^2 = [1, -3, 9]^T S [1, -3, 9]^T \approx 5.25$$

$$\therefore p(f^*|y, x^* = -3) \sim \mathcal{N}(f^*|2.347, 5.25)$$

The post. pred for the label is $p(y^*=1|y, x^*) = \int \sigma(f) p(f|y, x^*) df$, which has no closed form for the logistic link.

Using probit approx: $\sigma(z) \approx \Phi(kz)$, $w/2 = \sqrt{\frac{2}{\pi}}$

$$\Rightarrow p(y^*=1|y, x^*) \approx \Phi\left(\frac{2 \mu_F}{\sqrt{2 \sigma_F^2}}\right) \Leftrightarrow \Phi\left(\frac{\mu_F}{\frac{\sigma_F}{\sqrt{2}}}\right), \quad \text{plugging in the numbers } p(y^*=1|y, x^* = -3) \approx \Phi\left(\frac{2.347}{\sqrt{2.625}}\right) = \Phi(0.84) \approx 0.8$$

$$\therefore p(y=0|y, x^* = -3) \approx 0.2$$

$$p(y^*=1|y, x^*) \approx \text{Ber}(q^*(p)) = \text{Ber}(q^*(0.8))$$

Consider the following utility matrix for a decision (i.e. the prediction) $\hat{y} \in \{0, 1\}$ and the true value $y \in \{0, 1\}$:

$U(y, \hat{y})$	$\hat{y} = 0$	$\hat{y} = 1$
$y = 0$	2	1
$y = 1$	1	2

Suppose the predictive posterior distribution for a specific x^* is given by $p(y^* = 1 | y, x^*) = 0.129$.

Question 3.5: Compute the expected utility for each decision $\hat{y} \in \{0, 1\}$ and determine the optimal decision wrt. the utility matrix above.

let's define

$$\begin{aligned} \bullet p &= P(y=1) \\ \bullet 1-p &= P(y=0) \end{aligned}$$

$$\begin{aligned} EU(\hat{y}=0) &= U(0,0) \cdot P(y=0) + U(1,0) \cdot P(y=1) = 2(1-p) + p \\ EU(\hat{y}=1) &= U(0,1) \cdot P(y=0) + U(1,1) \cdot P(y=1) = (1-p) + 2p \end{aligned}$$

More general

$$EU = (y) = \sum_{y \in \{0,1\}} P(y) \cdot U(y, \hat{y})$$

- plugging in the numbers: $P(y^*=1 | y, x^*) = 0.129 \Rightarrow P(y^*=0 | y, x^*) = 0.871$

$$EU(\hat{y}=0) = U(0,0) \cdot P(y=0) + U(1,0) \cdot P(y=1) = 2(1-p) + p = 2(0.871) + 0.129 = \underline{\underline{1.871}}$$

$$EU(\hat{y}=1) = U(0,1) \cdot P(y=0) + U(1,1) \cdot P(y=1) = (1-p) + 2p = 0.871 + 2(0.129) = \underline{\underline{1.129}}$$

we choose $\hat{y}=0$ as it maximizes utility

Task 5.3 Estimate the post. mean of $\sin(2_1, 2_2)$ using the samples

Like normal expectation

$$E_p[f(x)] = \int f(x) p(x) dx$$

- the posterior mean can be written as $E[\sin(2_1, 2_2) | D] = \int \int \sin(2_1, 2_2) P(2_1, 2_2 | D) d_{2_1} d_{2_2}$

- using MC samples, we estimate the mean $E[\sin(2_1, 2_2) | D] \approx \frac{1}{S} \sum_{i=1}^S \sin(\underbrace{2_1^{(i)}, 2_2^{(i)}}_{\text{posterior samples}}) \approx 0.13$

posterior samples: $2_1^{(i)}, 2_2^{(i)} \sim p(2_1, 2_2 | D)$

Task 5.4 Estimate the posterior prob $p(2_1 > 2_2 | D)$ using samples

$$p(2_1 > 2_2 | D) \approx \frac{1}{S} \sum_{i=1}^S \mathbb{1}[2_1^{(i)} > 2_2^{(i)}] \approx 0.68$$